

infidex 176



Камера
ПАНОРАМНАЯ

Printing and Assembly guide

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S *steps*

Four steps for successful assembly. Each step is described in detail.

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INTRODUCTION

Infix 176 - is the open source project of 3D printed panoramic film camera. The project is designed to be printed on an FFF printer, so if you plan to print using a different technology, I cannot guarantee the same result.

It is important to understand that this is a completely ready-to-use project, but it still provides plenty of space for your needs and creativity.

And it is even more important to remember that this is a plastic camera. What imposes some restrictions. Plus, let's not forget that your printer, the chosen plastic, finishing materials and the number of stones in your neighbor's garden play a huge role (about 80 percent). By this I mean that it may take more than one attempt to turn a roll of plastic into a working camera. But I ask you not to dwell on failures, but to analyze and overcome problems.

I printed all my cameras on a stock Ender 3 with Marlin firmware and Linear Advance enabled using cheap ABS plastic. So I can't guarantee that all the tolerances and scales will work for your setup. Experiment.

MATERIALS

List of necessary materials and tools

- Mamiya TLR system taking lens (C2(Mamiyaflex)/22/220, C3/33/330)
- 1 roll of filament (whole camera takes less than 600 grams - 22 ounces)
- Superglue (preferably two-component with an activator, but I used regular one)
- Rigid wire with a diameter of 0,6-0,8mm (20-30cm will be enough. I used stainless steel wire. It should spring well - this is important)
- Black velvet self-adhesive paper (flocked paper) or vinyl self-adhesive film
- Black foam (at least dark gray)
- Double sided tape (not foam based)
- Pair of nippers for wire
- Calipers with metric system
- Sandpaper (depends on the material you are printing with, take from 100P up to 800P)
- Spanner wrench
- Scissors
- Tweezers
- Protective goggles and respirator

List of optional materials and tools

- 35mm optical viewfinder
- Cutting knife (with changeable blades is fine)
- 3d printing adhesive glue (i used in spray format and can recommend to use it for you too)
- Epoxy glue
- Putty for plastic
- Light grain sandpaper (1000P)
- Texture car bumper paint ("Mipa" for example)
- Brass heat inserts with 1/4" thread 8mm x 8mm
- Brass heat inserts with M3 thread 4,5mm x 6mm (and at least four M3*10 screws)
- Soldering iron

Lens

As you can probably guess at this point, project requires the taking lens from Mamiya TLR system. This system has different focal lengths (65mm, 80mm, 105mm, 130mm etc). In my opinion 80mm works the best. It provides 35mm equivalent focal length on 72mm panoramic frame (in width). Which allows you to use 35mm optical viewfinder. Also it's really easy to find and buy even with a whole Mamiya camera and you don't need to mess up it since the lenses on this system are interchangeable. Lens is holded by a thread nut which you need to hold lens on the infidex camera. Unfortunately you need to desolder a synchro contact wire.

About Lomo T-22 lens: you still can attach it but only with the cone from previous IV version of infidex. Files are included

Filament

I prefer printing all my cameras using **ABS** for specific reasons: it is easy to shape with hand tools and believe me, there will be a lot of shaping; it is cheap; it has good mechanical properties for use in various environments. However, the material is quite finicky and shrinks during printing and cooling; requires adhesive glue; slow speeds (from 30mm/s to 70mm/s max); enclosed chamber and full cooling down awaiting till you can pick up parts, unless they're gonna delaminate.

My friend printed a few cameras out of **PLA** and they work absolutely fine. **PETG** can be too flexible in some parts. **Composite filaments** good at hiding printing imperfections, looks great and in theory they should be perfect for this role, but not every printer can handle composites, plus they are really expensive.

Other materials and tools will be described as you progress through the instructions.

Precautionary measures:

Please wear respiratory and eye protection when handling toxic plastics such as ABS and when working with superglue. When working with a cutter knife, do not point the tip of the blade in your direction. Work carefully, without rushing.

PRINTING

Choosing parts for your camera

The fifth version of the camera offers some choice of parts for greater ease of use and modifications. Read the manual to the end to see all the options available for printing.

The order in which the parts are printed does not matter. However, I recommend printing the body first to check the measurements and tolerance, if necessary you can later scale the parts and reprint them. Some reasonable uncertainty in measurements is possible.

Common printer settings

Here is list of **my settings** which i used to print all my cameras with ABS plastic:

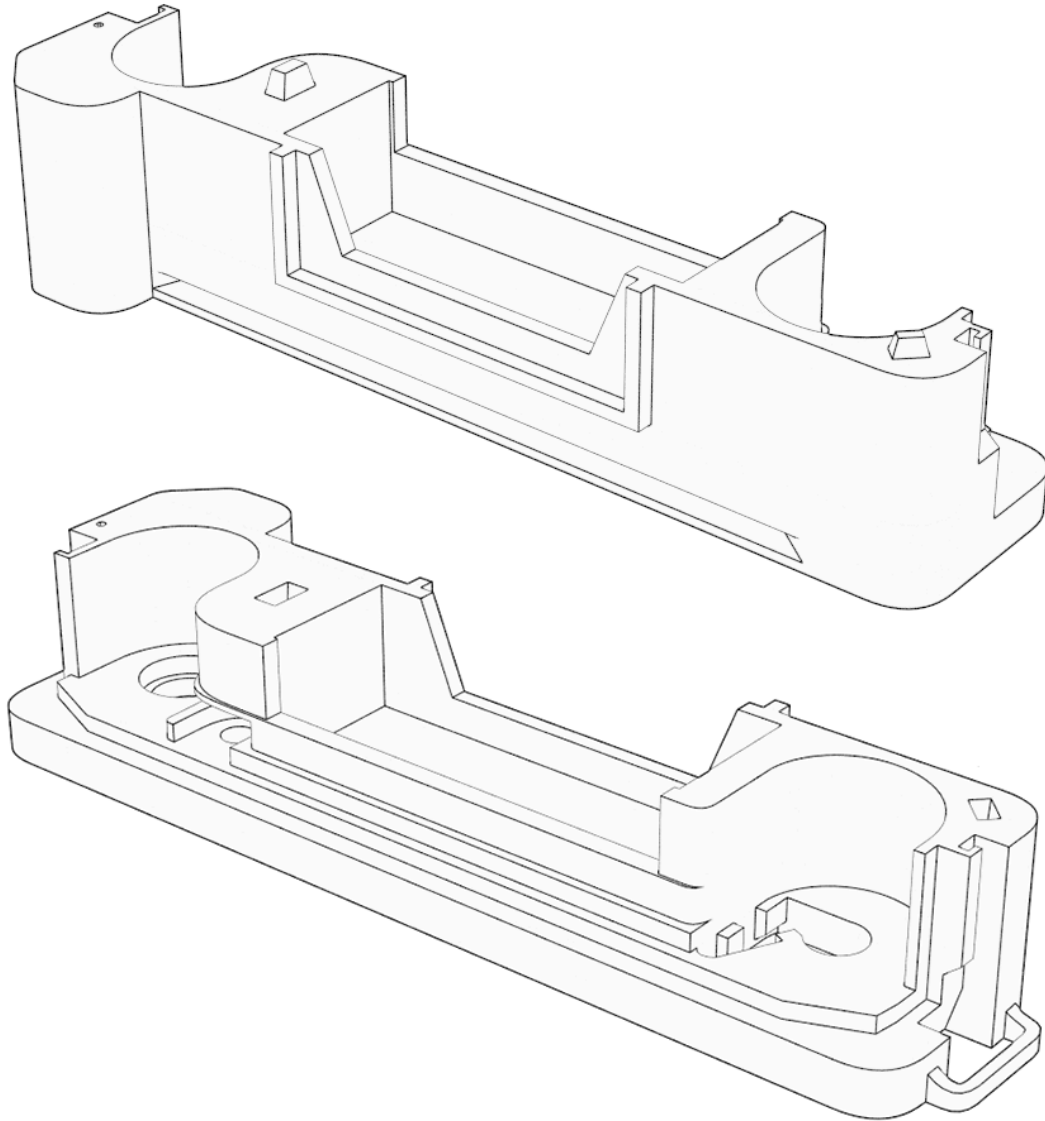
Nozzle	0,4mm
Layer height	0,2mm
First layer height	0,23mm (depends on your bed. Mine is glass)
Number of inner perimetrs	4 (an important parameter ensuring rigidity)
Minimal thickness of top and bottom layers	2mm
Infill	25% max (shape on your taste)
Bed surface	4mm glass (normal polished glass) Of course PEI and other works fine

BODY

VARIATION 1

Contain two parts: **BODY_bottom_var1** and **BODY_top_var1**

Supports - Yes, “from bed” mode. I used “tree” supports on every part



placement on printer bed

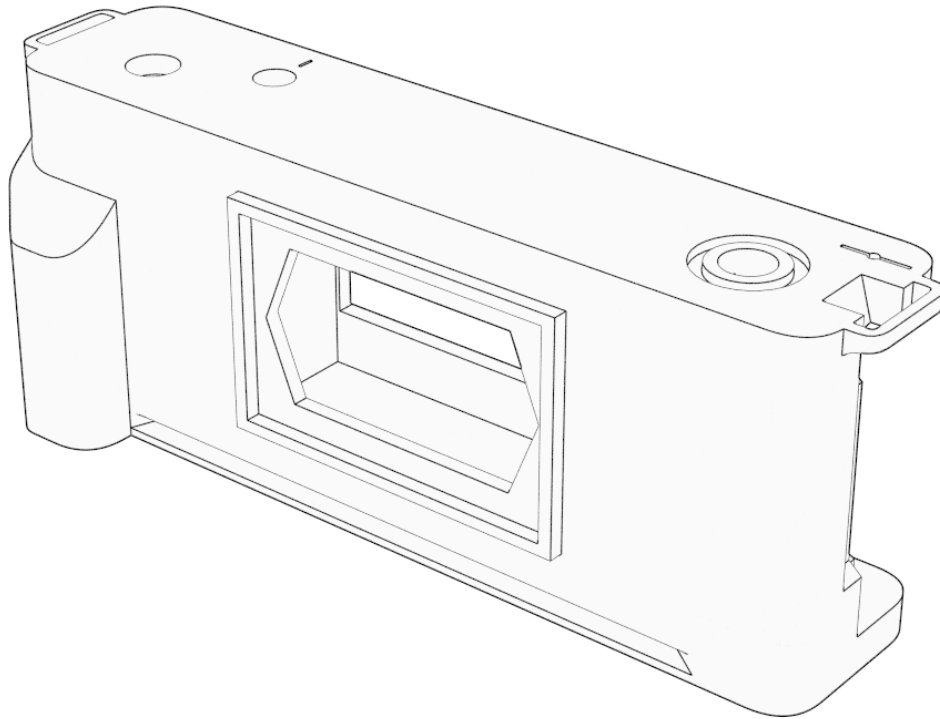
VARIATION 2

Contains same two parts **BODY_bottom_var2** and **BODY_top_var2** with additional **holes in front** for M3 brass inserts for only mounting helicoid. Same placement.

Supports - Yes, “from bed” mode

VARIATION 3 - "Unibody"
Contains one model **BODY_unibody_var1**

Supports - Yes



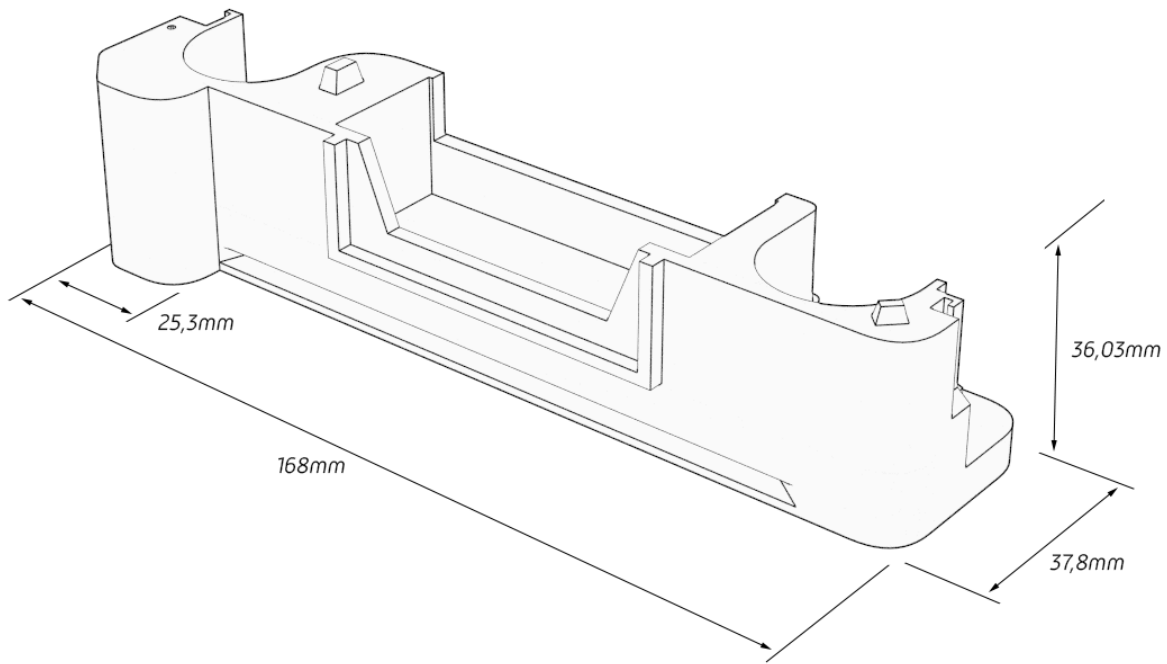
probably placement on "top to down" should be better (need tests)

VARIATION 4 - "Unibody"
Contains one model **BODY_unibody_var2** with additional **holes in front** for M3 brass inserts for only mounting helicoid. Same placement.

Supports - Yes

CHECKING THE DIMENSIONS

After printing the bottom body part (and before printing, in slicer) check the dimensions with reference on the next page. Minor and reasonable uncertainty in measurements is possible. About 0,2-0,4mm is absolutely fine (not in slicer of course).



full height of body ("unibody" for example) is 67mm

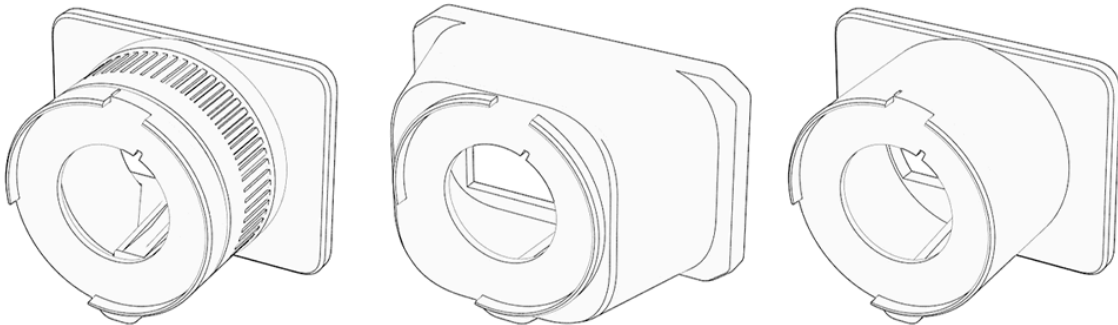
LENS MOUNT

3 variants of lens mount are available:

Focusing helicoid

Classical cone

Tube mount

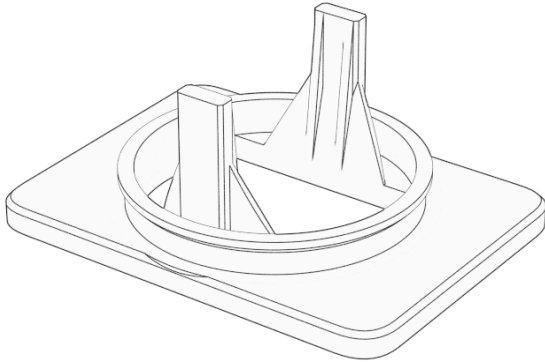


Depending on your choice you need to print only one set of parts. Also the lens cone from previous IV model is fully compatible with new V model. And with slightly modification of ledge on your IV model body you can attach this new helicoid lens mount too. All three available with M3 screw holes.

HELICOID VERSION LENS MOUNT

HELICOID MOUNT VARIATION 1

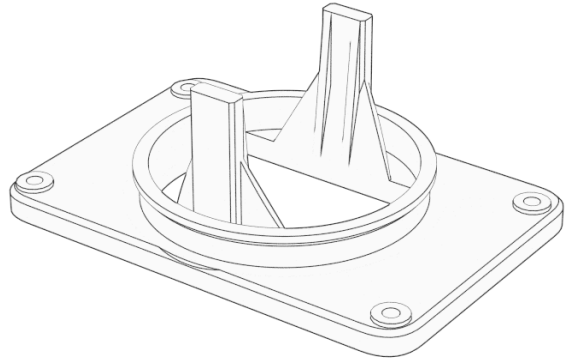
Contain: **Helicoid_MOUNT_ver1**



placement on printer bed

HELICOID MOUNT VARIATION 2

Contain: **Helicoid_MOUNT_ver2**

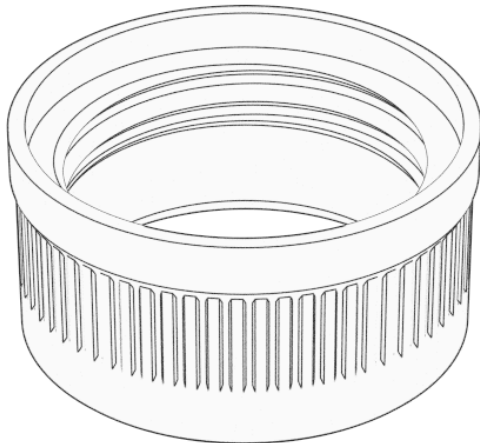


placement on printer bed

Helicoid mount version with four holes for attaching it to the body with four M3*10 screws.

HELICOID FOCUS RING VARIATION 1

Contain: **Helicoid_FOCUS_ring_ribs**

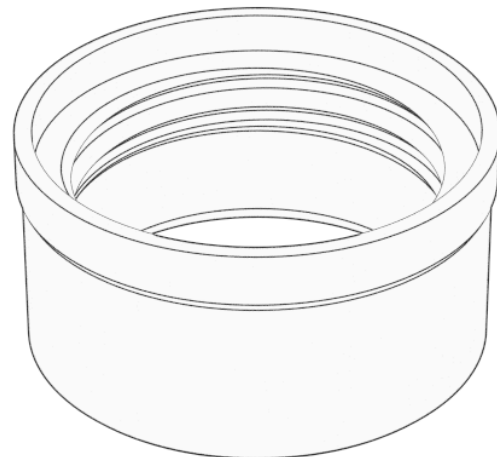


placement on printer bed

HELICOID FOCUS RING VARIATION 2

Contain:

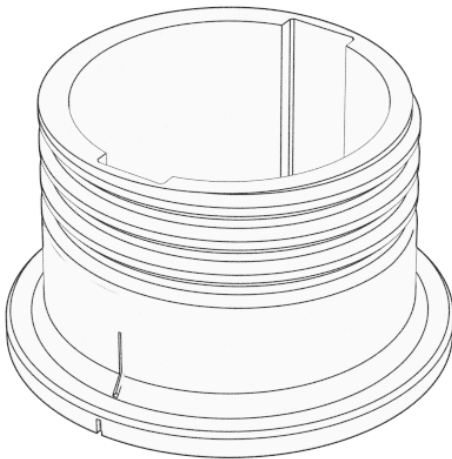
Helicoid_FOCUS_ring_NO_ribs



placement on printer bed

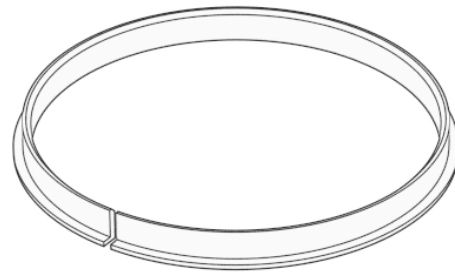
Same focusing ring just without ribs.

HELICOID INNER RING
Contain: **Helicoid_INNER_ring**



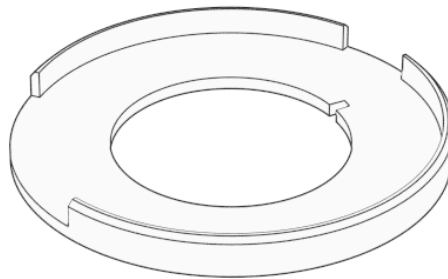
placement on printer bed

HELICOID FIXATOR RING
Contain: **Helicoid_FIX_ring**



placement on printer bed

HELICOID LENSE BOARD
Contain: **Helicoid_Lense_BOARD** and **Helicoid_Lense_BOARD_lower***

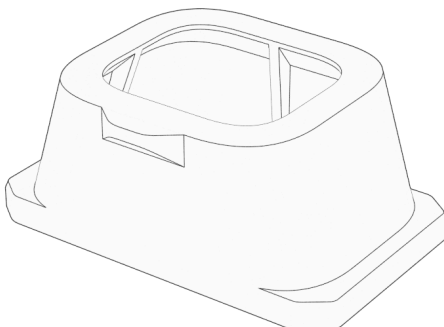


placement on printer bed

**lower version for focal length compensation. Start with normal one*

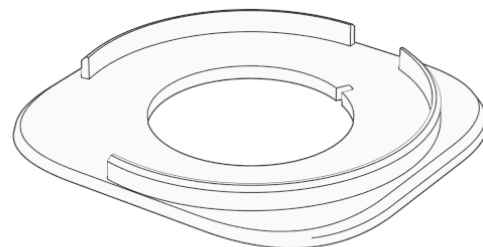
CLASSICAL CONE LENS MOUNT

CONE BASE
Contain: **CONE_lens_classic_BASE**
also available with M3 holes



placement on printer bed

CONE LENSE BOARD
Contain: **CONE_lens_classic_BOARD**



placement on printer bed

IMPORTANT MESSAGE FOR “BLUE DOT” USERS

After quite huge investigation that my friends did, it was established that “Blue Dot” lenses do not share the same flange distance as regular 80mm lenses. I started receiving messages from different people, who had a copy of the Blue Dot lens, that they have difficulties with achieving the infinity focus. It was always the blue dot version. My friend Giuseppe found information about its unique characteristics online. After that i designed a short version of helicoid for @analog_astronaut and he confirmed that now he has infinity focus. Of course he made some stunning images as proof.

What does it mean to you? Well, if you you have blue dot lens, you have to print following set of parts:

1. **Helicoid_INNER_ring_80mm_shorter-2mm.STL**
2. **Helicoid_FOCUS_ring_ribs_shorter-2mm.STL** (or without ribs on your choice)

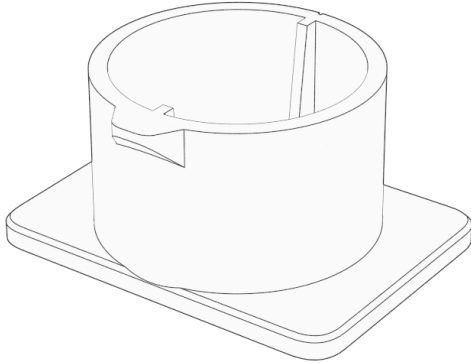
New files are included.

Huge thanks to Giuseppe who found that information and to the @analog_astronaut for boosting up this project in the media. You guys rocks!

TUBE LENS MOUNT

TUBE MOUNT

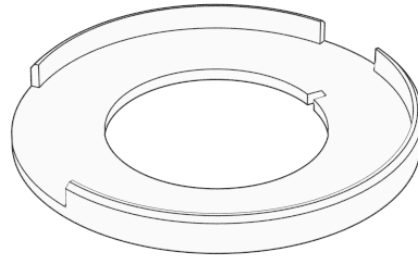
Contain: **Tube_lens_mount**



placement on printer bed

TUBE LENS BOARD

Contain: **Tube_Lense_BOARD**

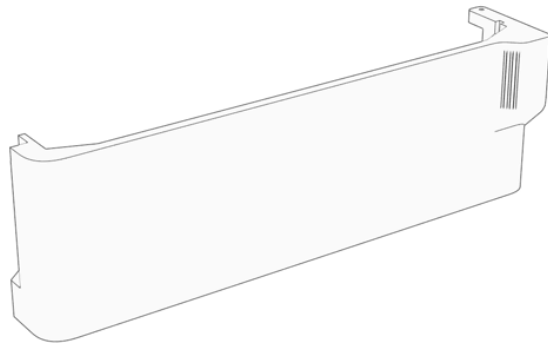
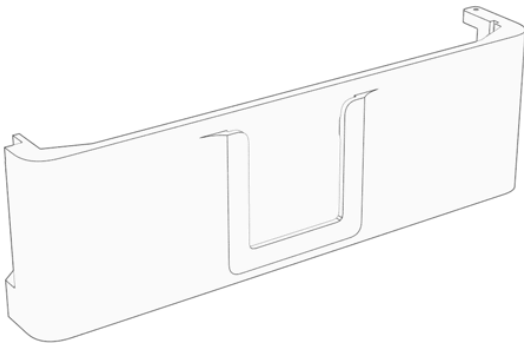
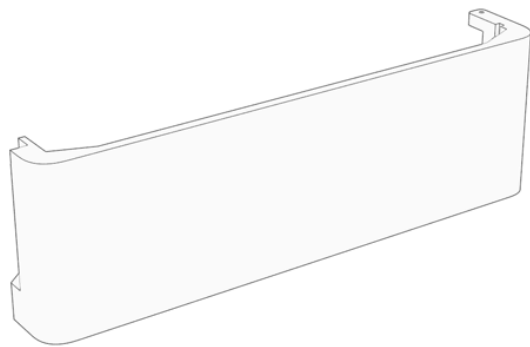
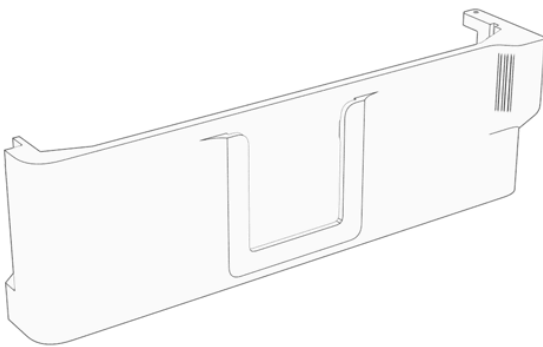


placement on printer bed

DOOR

Four versions available. It's simply the door with or without film reminder and thumbrest in different combinations. Use rafts/brims if necessary.

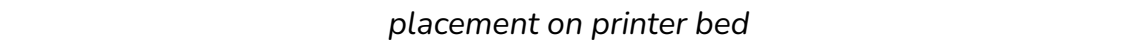
Supports - Yes, "from bed" mode



placement on printer bed

NOTE: the lower ledge on the door is gonna sag a bit because of the supports , so you need to remove some material to make it fit into the door frame.

n) separately to achieve



nd viewfinder are available

ASSEMBLY

Checking the fitting of parts

Before you can permanently glue your fingers to freshly printed parts, let's check how they fit together. That's the step where you need your optional tools such as file or sandpaper, cutting knife etc.

In general - you should check the fitment of big parts first. Too loose - reprint. Too tight - shape it with tools. Rotating parts should not get stuck or dangle (the exception is the taking spool shaft. The seat-hole at the bottom of the body is made so that the shaft can go slightly deeper into it).

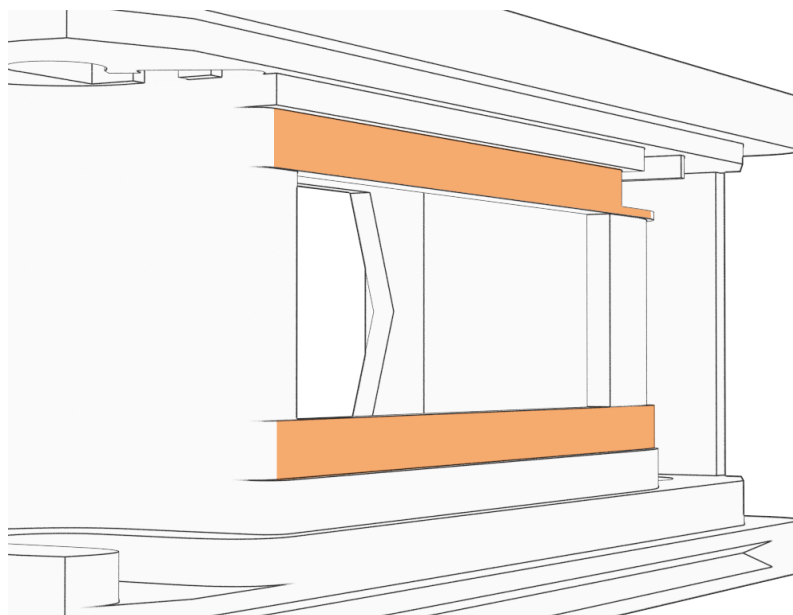
ATTENTION! The glue fumes are dangerous and will distract you from gluing as your eyes will water. Please wear safety glasses and a respirator!

BODY

“Unibody” dudes can move to the next step.

Two halves of the body probably have a gap between, and if it's not wider than 0.4mm - you're good to go. Just need some sanding and more pressure while you are gluing them, probably a good idea to use clamps. And of course you need to fill this gap with putty or solder it.

Critical is to make frame planes as parallel to each other as possible, since it will affect your focus plane. You can make it by placing both parts on a table, or other flat surface, or use film pressure plate as a guide by placing it on the orange film rails.

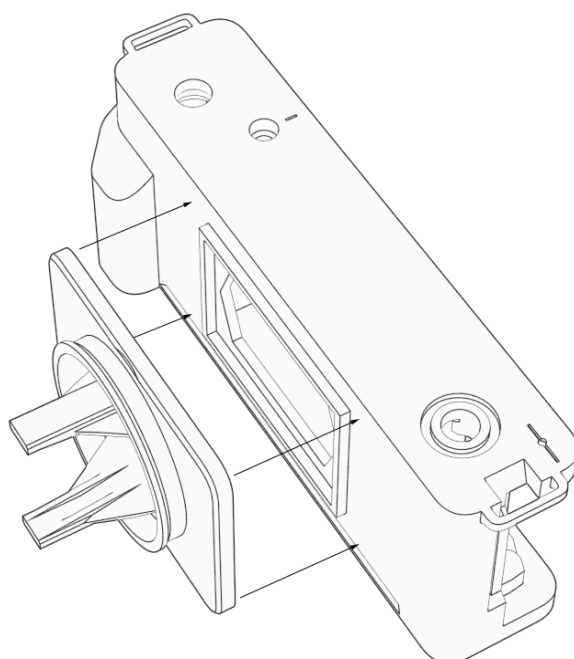


If you chose a version with holes for brass inserts, it's time to insert them with soldering iron. Please be careful. Holes are for **M3** brass inserts (diameter - **4,5mm**; height - **6mm**). Don't pull your iron away, there is a hole on the bottom of the body for $\frac{1}{4}$ " (diameter - **8mm**; height - **8mm**) brass insert, so you can use other tripods platforms.

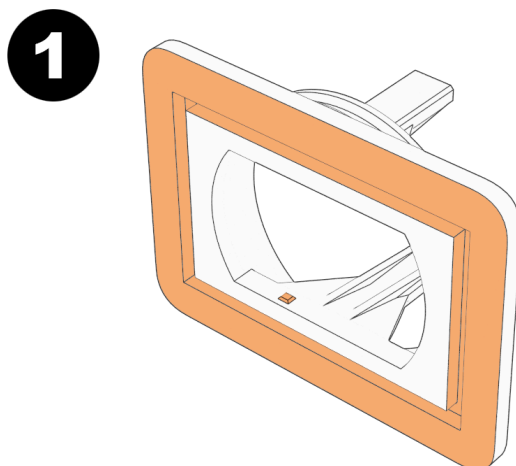
NOTE: after installing brass inserts some excess plastic will be squeezed out. You need to remove it flash with the surface of camera body.

LENS MOUNT

After gluing the body, you can check the fitting between it and lens mount, installing it in the ledge on the body. If necessary, grind some material off the lens mount part.

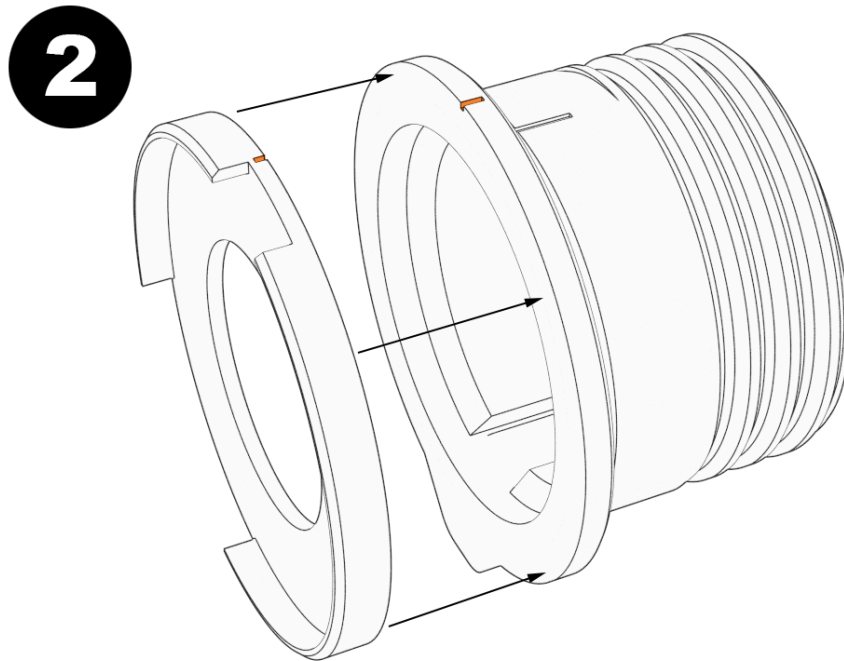


1. Once you are sure that everything is ok, glue the lens mount to the body using a superglue or epoxy glue. Or temporary attach it with M3*10 screws. Take a look at the small box on the part (highlighted orange) that indicates the bottom side. Highlighted in orange - is the place you should apply glue (but not the small box of course 😊).



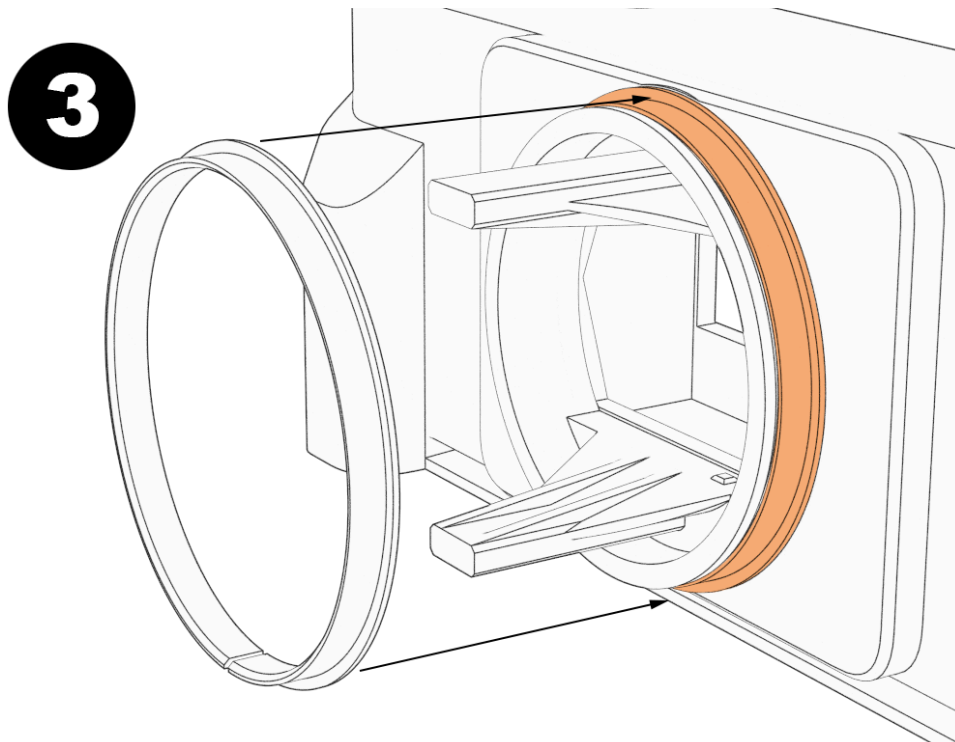
Tube mount and classical lens cone have the same method of installation.

2. Next step is to glue the lens board on the inner thread ring. Check the highlighted marks, they should be on one line. **Lower version** of **lens board** needs when focal length is too long for infinity focus.

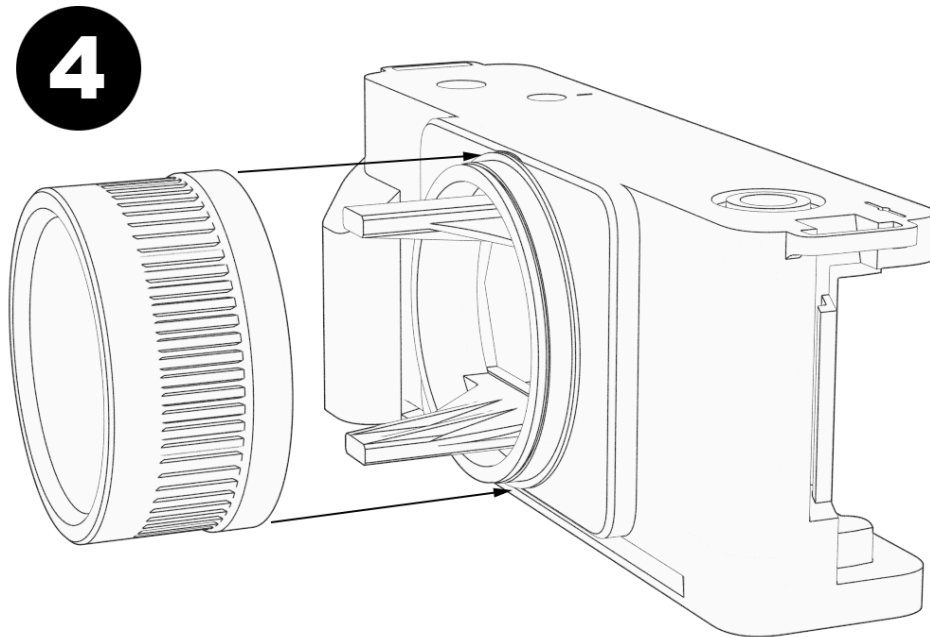


Tube mount and classical lens cone have the same method of installation of lens board.

3. Now snap the fixation ring on the lens mount groove. Orange highlights the spot for placement position.

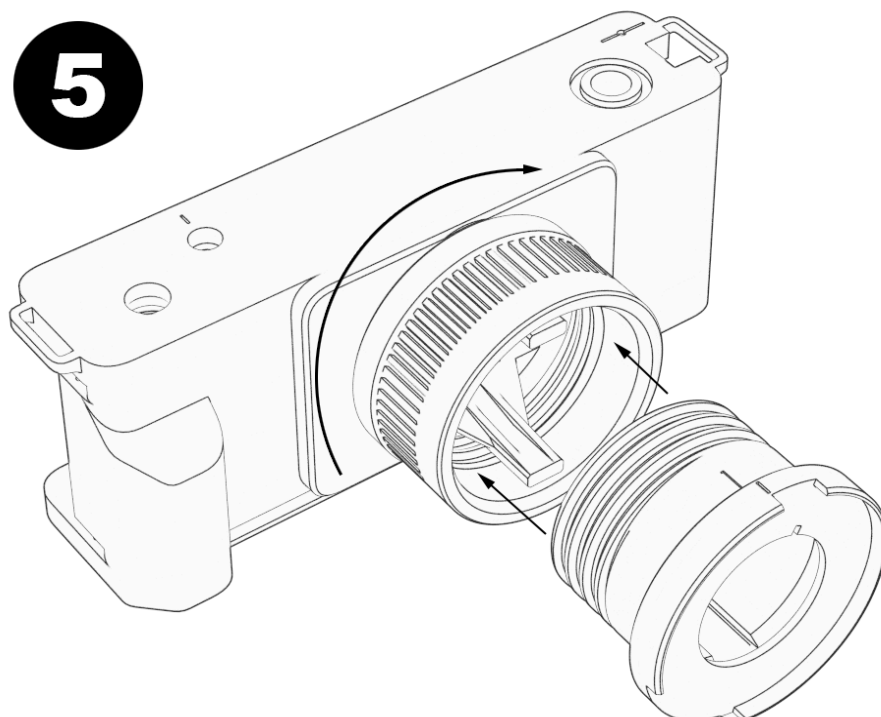


4. Place glue on the surface of the fixation ring that we just snapped in place. **ATTENTION!** Be really careful not to glue it to the lens mount groove. Then place the focusing ring on that fixation ring and press it mildly for a few seconds. Check the rotation. It should rotate freely without dangling and without the possibility of falling out of the groove.



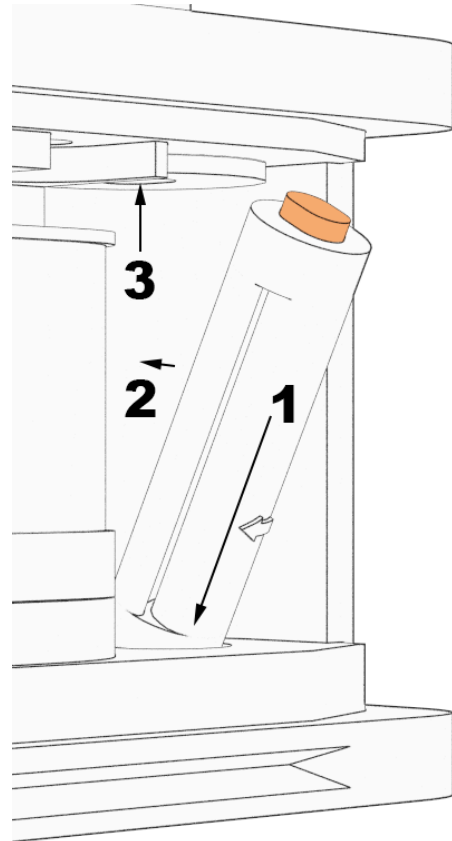
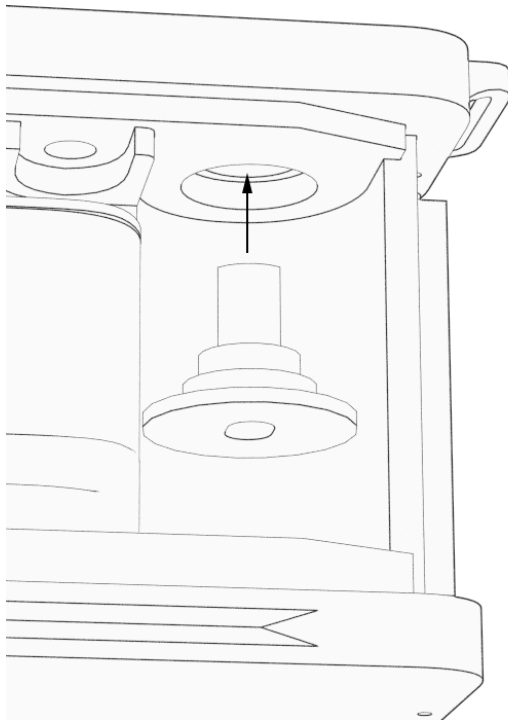
5. Check the inner thread with lens board operation. Place it inside the focusing ring so that the slides of the mount fits into the grooves of the inner thread ring. Then simply start rotating the focusing ring clockwise. It should screw in all the way.

If it takes too much force to screw it down, probably you need to print smaller version of the inner thread, or sand surface a bit. If you are planning to grease it, don't forget to use thick grease instead of liquid.

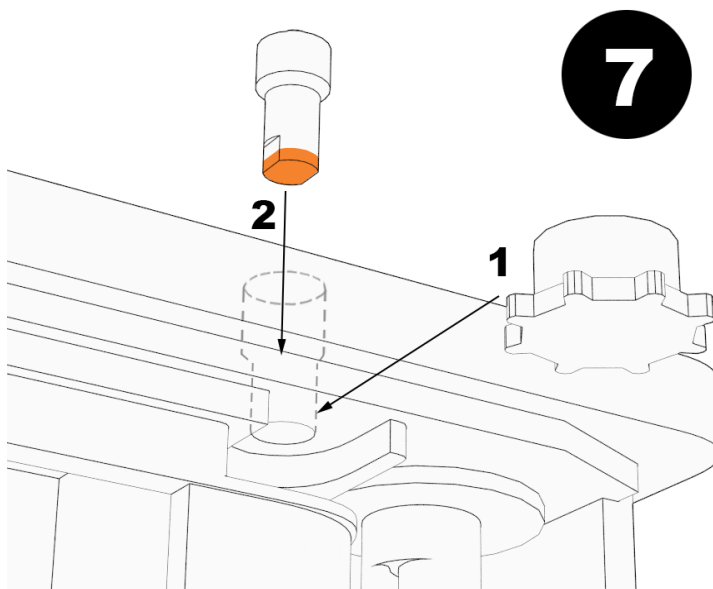


After checking, you should unscrew it back, as it may interfere with further assembly.

6. Place the spool base part on the top side of the body. Then apply glue on the highlighted part of the spool shaft, and using tweezers, place it inside of the camera body at a slight angle, push it down and forward, until it stops. Then glue together two parts. Check the rotation. This part should have free space for moving down and up for now. You should glue the winding knob at top after you finish body and paint works.



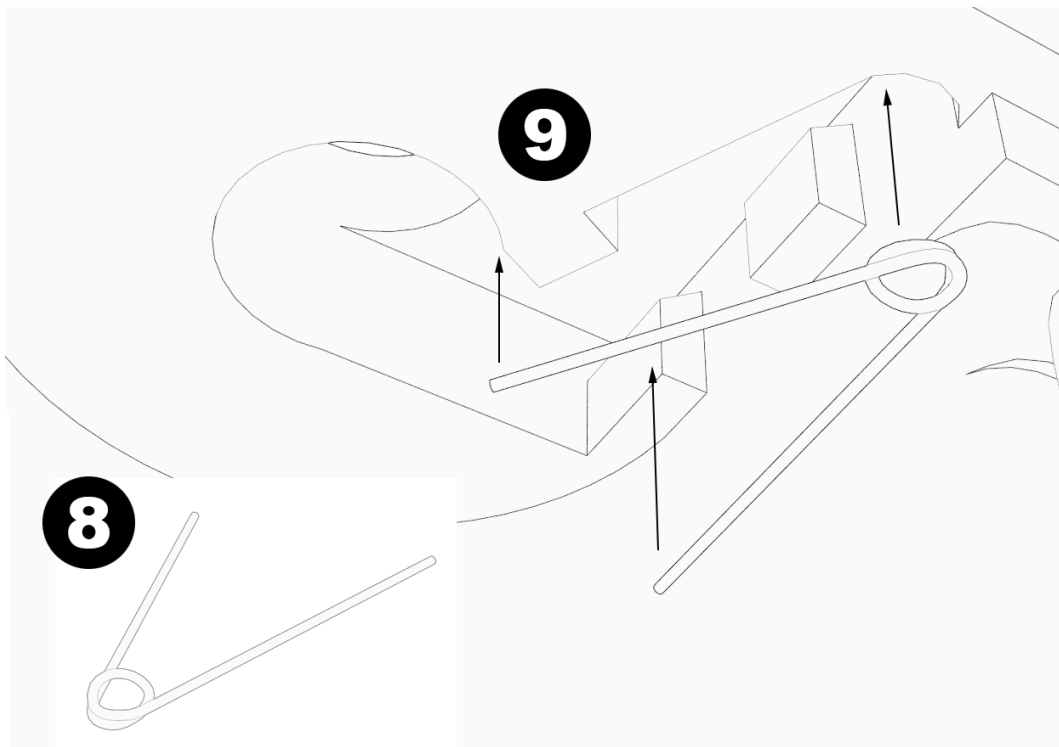
7. Place the sprockets gear into the slot, then apply glue to the sprocket shaft and attach it to gear through the shaft hole from up side.



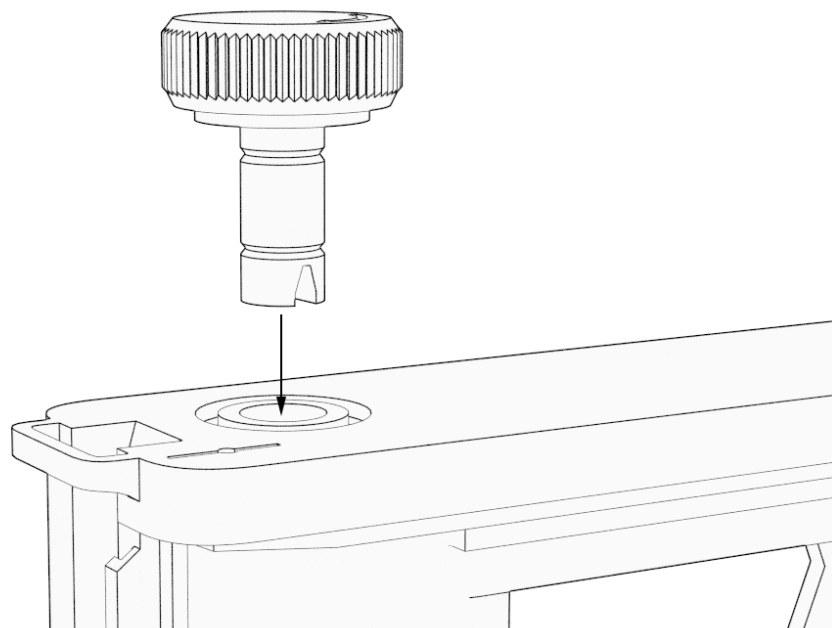
After you finish a paint job, you should glue the frame counter cap to the sprockets shaft. Of course don't forget to mask these small parts on top, otherwise you will glue them to the body with paint (or simply place it on after painting).

8. Now you need to bend the wire into a V-shaped spring. I recommend doing this with a small coil so that it is stiff enough to fix the rewind knob. Take the measurements and cut it.

9. After, place it inside of the slot. There are three bumps where it should be installed.



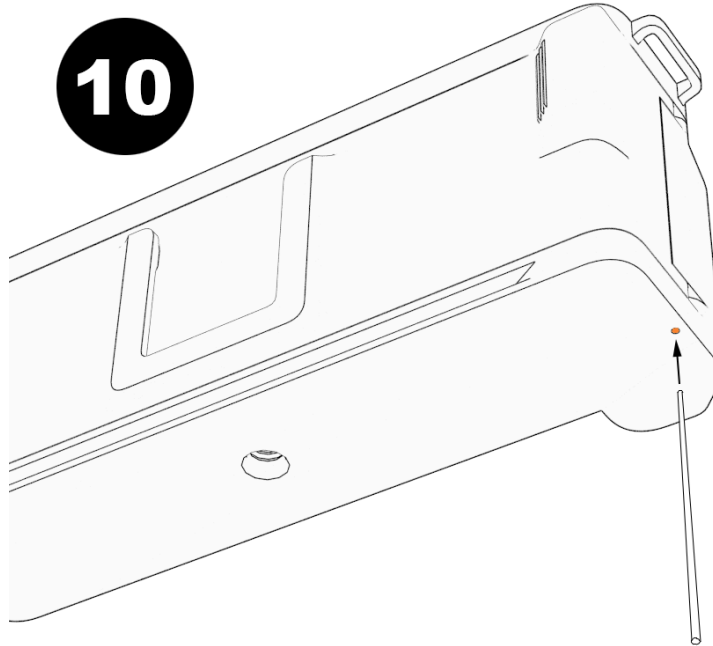
Next, insert the rewind knob and make sure it locks in the closed and open positions. After checking, better to pull it out, as it may interfere with further assembly and painting. **These steps are tricky, be patient, I believe in You.**



DOOR

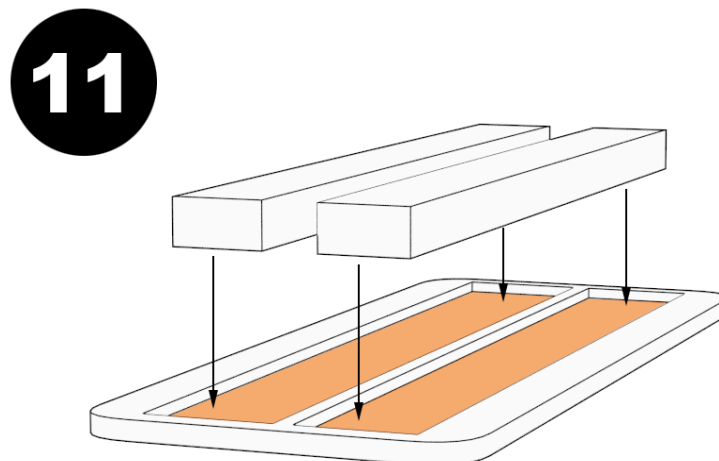
10. Next, we will install the door with the film pressure plate. First place the door. Check its fit. Door can be a bit tight, that's ok. After you are sure that the door opens and closes, set it in the closed position and hinge it with the wire putting it through the hole in the body below to secure the door. The wire should fit tightly and not be able to fall out during operation. For reliability, this hole can be soldered or filled with putty.

Length of wire is about **63mm**

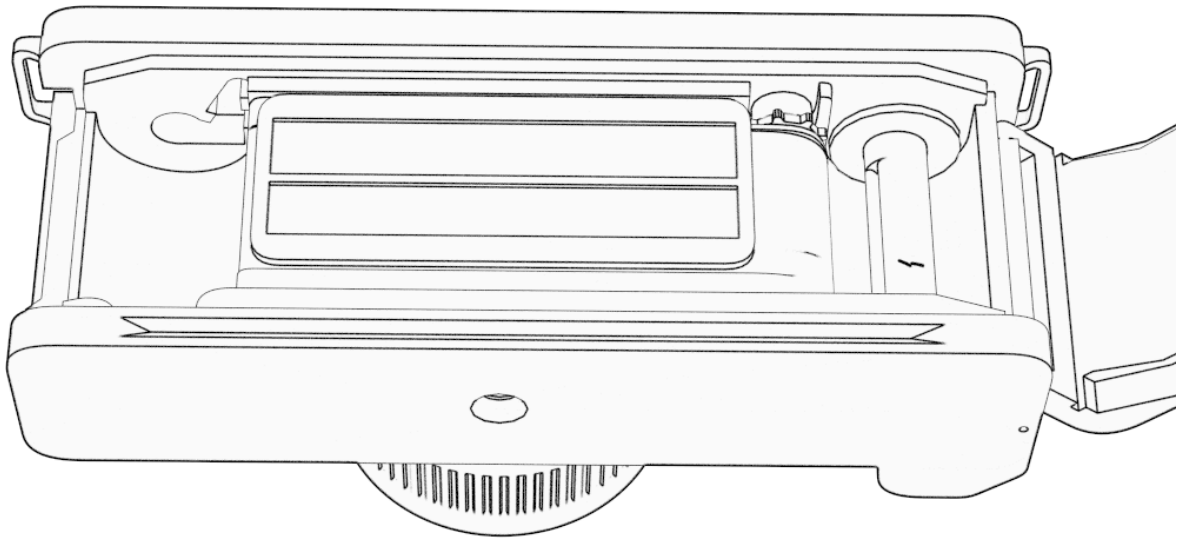


Next, I strongly recommend covering the flat side of the pressure plate with either black velvet self-adhesive paper (flocked paper) or matte black vinyl to prevent scratches on film. Do not use gray or even more light material unless you want strong halations and a hazy effect on your images. Templates are on the last page.

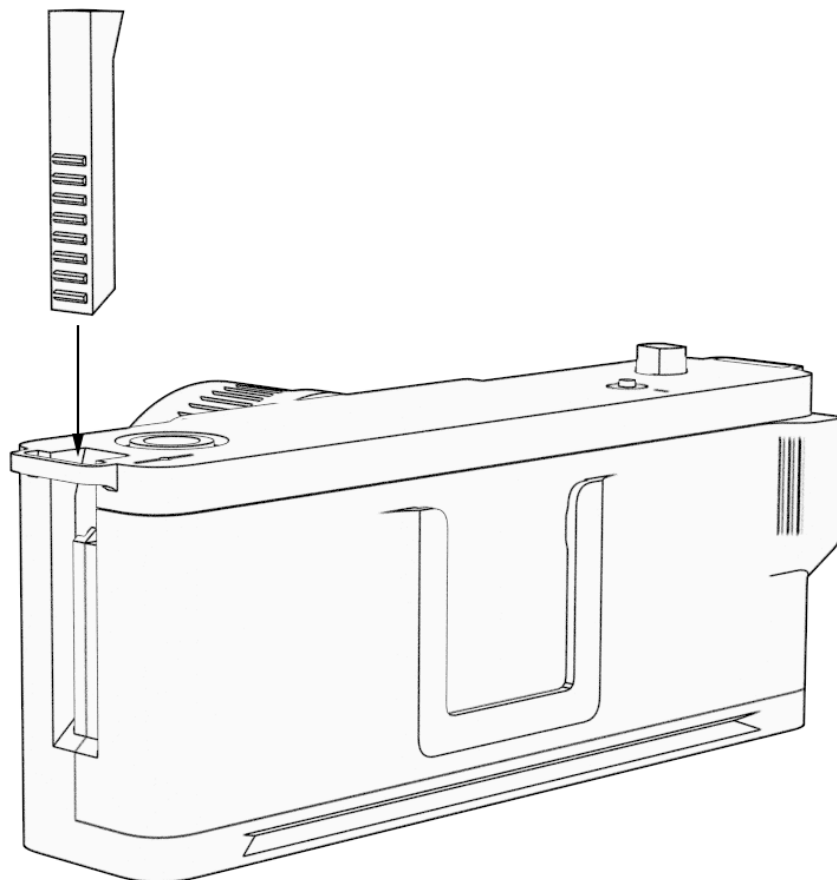
11. Then, take and cut two strips of foam, apply double-sided tape to them on both sides. Then attach these strips to the plate between the ribs. There is no need to make the foam too thick. 4-5mm is quite enough.



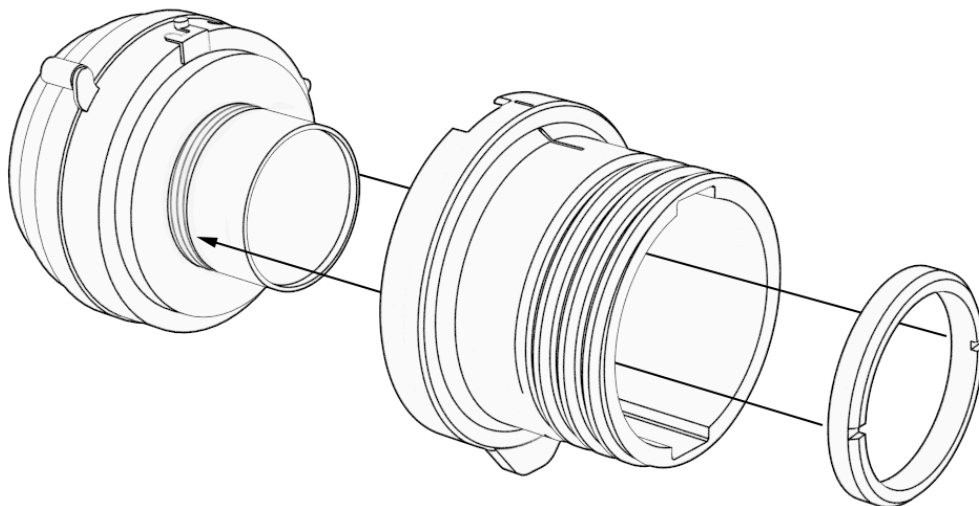
Then place the film pressure plate flat on the inside of the body, in place where an actual film is supposed to be, so that the plate does not touch anything inside and lies flat covering the frame window. Prepare double-sided tape and close the door completely.



After opening the door, the film plate should stain on it. Gently press it against the door a couple of times to make sure everything is ok. Take the latch and check how it fits into the lock place. Door should be flush with the surface of the camera body.

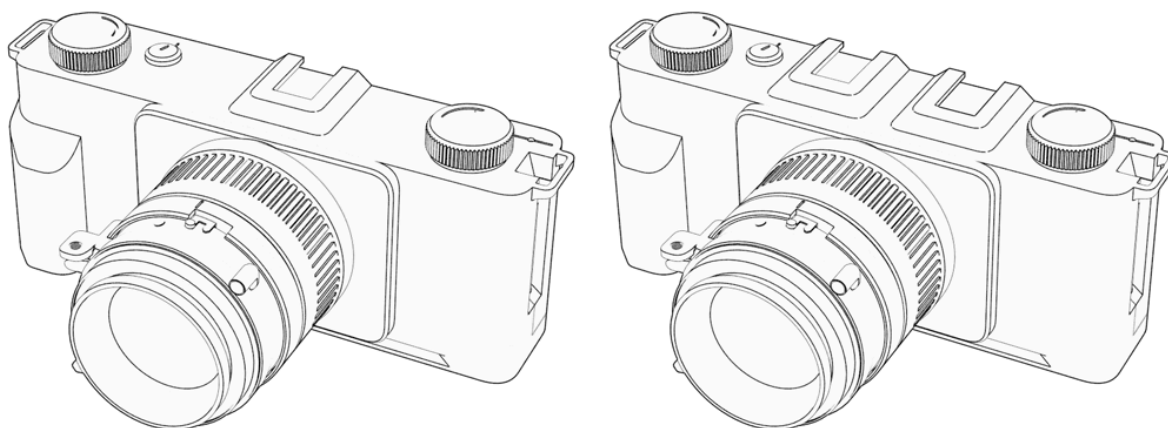


Finally attach the Mamiya lens to the inner thread ring using the original mount nut which was on the Mamiya TLR lens board.

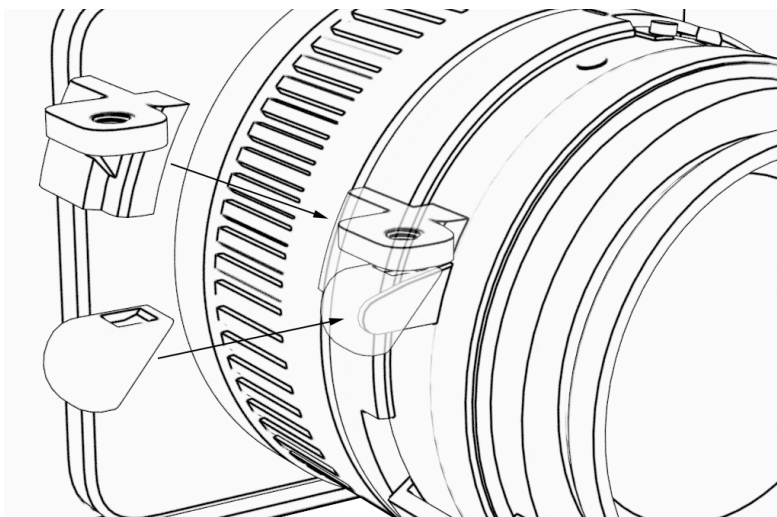


SMALL PARTS

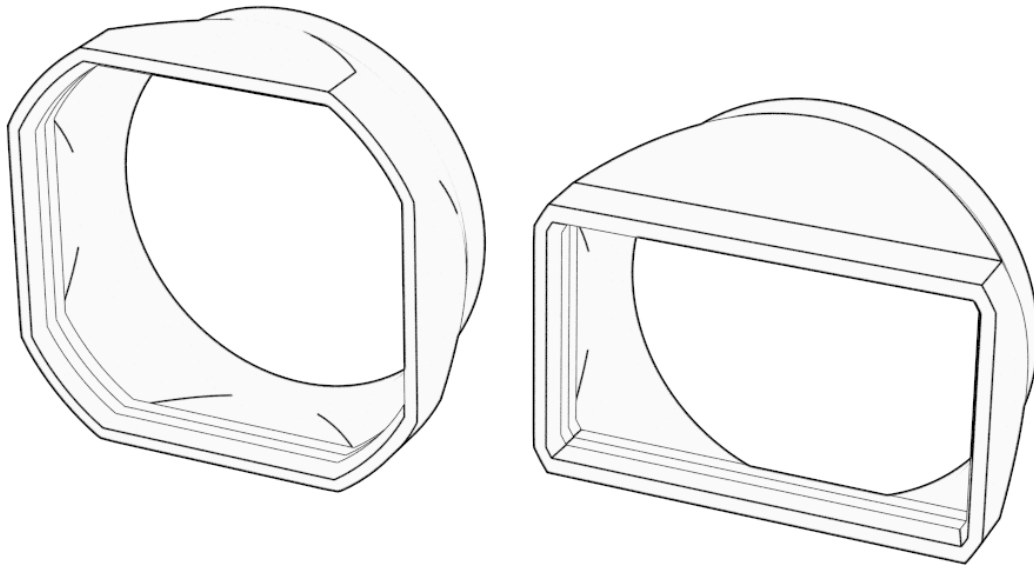
12. You can place cold shoe on preferred place. It could be a cold shoe from this kit, or a third party one.



13. Unfortunately Mamiya TLR lenses have no cable release thread port. So i recommend you to install it and lever knob too. Layer height is better to be about 0.1 - 0.12mm.



14. Also you can print lens hood, which is available in two variants. It is held on the front lens element by friction. Better to check your lens diameter before printing the hood and scale the hood to it.



FOCUSING

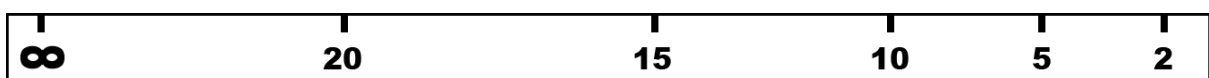
After all the necessary steps have been completed, you need to set the infinity mark on the focusing ring. To do so, you need some sort of ground glass. In my case I use 1mm thick glass for microscopes. It's conveniently 75 by 25mm. I used special matting paste for glass to make it frosted on one side.

Basically you need to screw the helicoid all the way in - that should be infinity focus, which on Mamiya 80mm is about 31 meters (or nearly 102 feet).

Then you wrap around the focusing ring piece of paper and make the marks on it for your preferred distances. My calculations show me that one 300 degree turn provides focusing from infinity up to 2 meters. Yes, you can go further and mark macro distances, **BUT be careful since the lens could fully unscrew and fall out.**

Infinity focus you can check simply by focusing on something really far away from your house. It could be a mountain, tower, tall building, you name it.

For focusing on near distances you need to measure preferred segments, such as 20 meters, 15, 10, 5 and so on. Place something bright and contrast on that distance and check via ground glass its focus. Make marks. Then scan this paper to make print of your own focusing marks.



*example of such strip of paper. length for 300 degree rotation is 170mm
it is an example, not an accurate thing.*

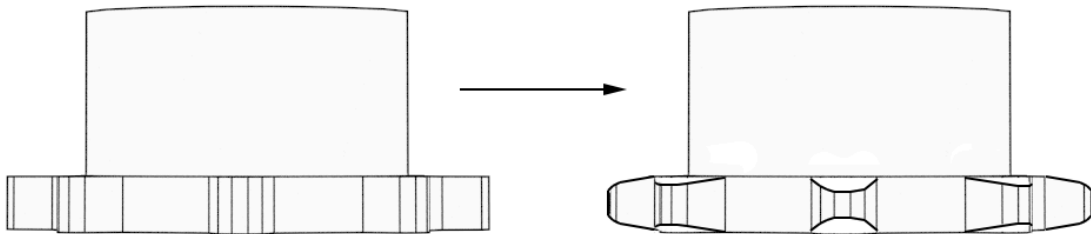
USAGE

First tests

Since you have already tested infinity focus and work of helicoid, the next test is advancing the film back and forward.

Better to have a 35mm roll of already exposed and developed film, or just really cheap or expired one, so that it wouldn't be a pity to ruin it. Simply try to imitate normal use of it.

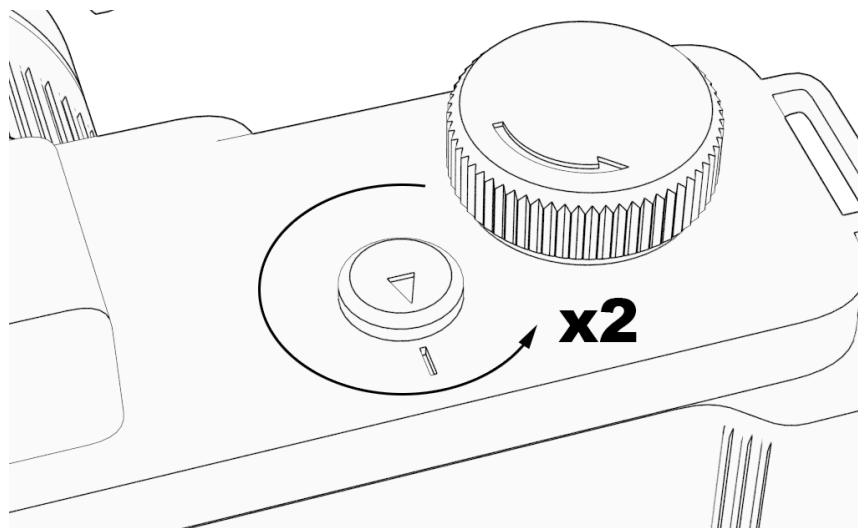
If the film is wound with a characteristic crunching sound and sprockets holes are torn - It is probably because sprocket gears are too sharp and square. In my case it was exactly like that. Simply just shape it with a file to smooth shape. You can copy the shape from other film camera if you have one.



It is also possible that your film will slip through a few gear teeth. If this happens, glue a piece of foam on the door next to a place where film ends contact with sprocket gears. It will provide more tension between gears and film.

Exploitation

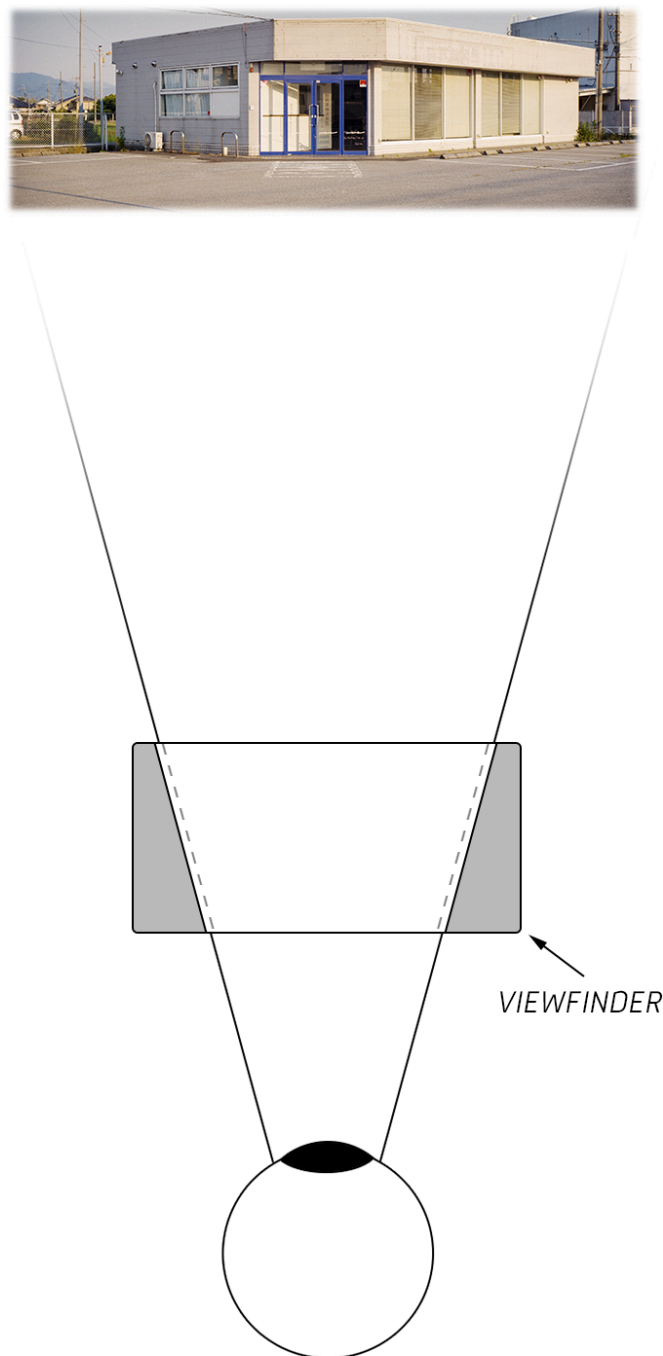
Main issue of infidex, which I still couldn't overcome, it's that you need to make 2 revelations of frame counter per 1 shot.



This problem can be easily fixed by replacing the 8-tooth sprocket gear with a larger 16-tooth gear. But it will increase the length of the camera body, which I really don't want.

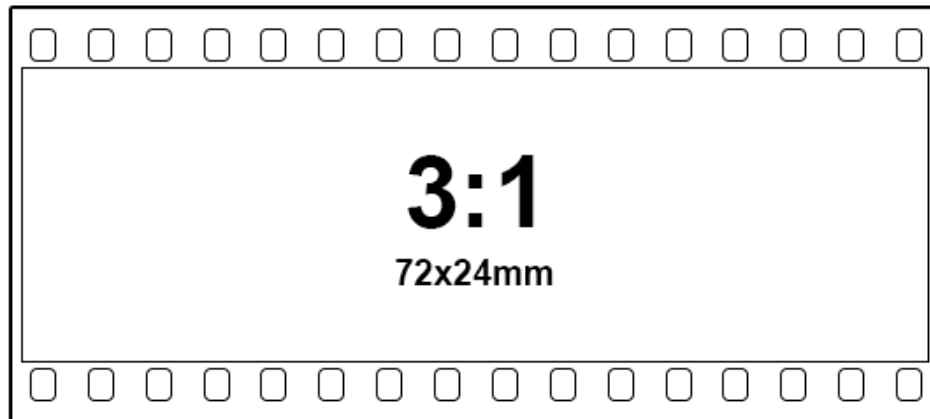
Viewfinder

Kit viewfinder is simple to use (which is available in two variations). just combine both frames into one silhouette, while viewing through, to get your correct framing. It is pretty accurate, but I recommend you keep some headroom on both upper and down sides of the frame. Or... just buy an optical 35mm viewfinder and mark new framing with ground glass, as I did.



Exposures

From one standard roll of film with 36 exposures, you get 18 frames if you do everything carefully, And 19 frames if you load in the complete dark and use the first frame without rewinding. However, this is a bit heavy and there is a chance that the film will not lie flat. Make sure that starting mark on counter is on start position.



One frame takes two standar 3:2 frames, so it's exactly 16 sprockets. Some respooled films can contain less than 36 exposures, keep that in mind.

Scanning film

Not any lab can properly scan such long negs, so I scan it on my own. I use the Epson V850 pro. This is the easiest way, but not the best (see example below). More interesting way is DSLR scanning, when you stitch two frames. But it is time consuming.



Kodak Gold, infidex 176 III with Lomo T-22

The end

I hope you will enjoy your new camera. Really want to see your images. You can send them to our joint gallery if you want: <https://timetowaste.ru/joint>

Peace.

Special thanks

Huge thanks to “PaperBen” for testing my designs, advice and inspiration. It really helps a lot man. I owe you a beer.

Big thanks to Giuseppe Spataro also for testing my designs, advice, inspiration and solving the infinity mystery for blue dot lens.

Thanks to “ballanux” youtube channel for inspiration on the helicoid system for the new V infidex 176 model.

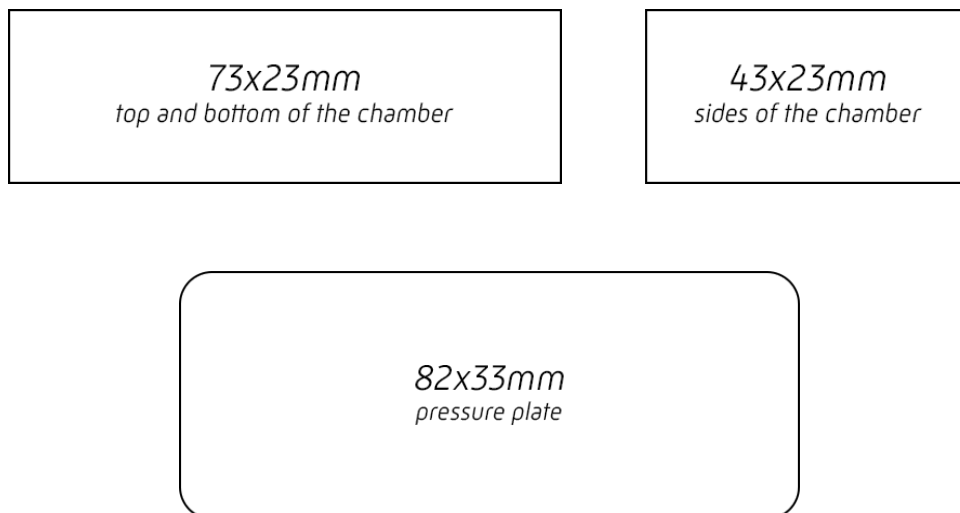
Thanks to Oscar Oweson (creator of Oxygen panoramic camera www.panomicron.com) for inspiration on some parts of the new V infidex 176 model.

Thanks to “GLOwl” (creator of printable lens cap www.thingiverse.com/thing:3241700) for inspiration on creating own version.

Thanks to the @analog_astronaut for boosting up this project in the media.

Templates

Templates for flocking paper



Depth of field table for Mamiya 80mm TLR

Aperture	Distance (metric)				
	inf.	9m	4,5m	3m	2m
2.8	32m - inf.	7m - 12m	4m - 5,5m	2,8m - 3,4m	2m - 2,2m
4	22m - inf.	6,5m - 15m	3,8m - 5,7m	2,6m - 3,5m	1,9m - 2,3m
5.6	15m - inf.	6m - 21m	3,5m - 6,3m	2,5m - 3,7m	1,8m - 2,4m
8	10m - inf.	5m - 51m	3,2m - 7,6m	2,4m - 4m	1,8m - 2,6m
11	8m - inf.	4,3m - inf.	3m - 10m	2,2m - 4,7m	1,7m - 2,8m
16	5,5m - inf.	3,5m - inf.	2,5m - 23m	2m - 6,4m	1,5m - 3,2m
22	4m - inf.	2,8m - inf.	2m - inf.	1,8m - 10m	1,4m - 4m
32	2,7m - inf.	2m - inf.	1,8m - inf.	1,5m - inf.	1,2m - 7,3m

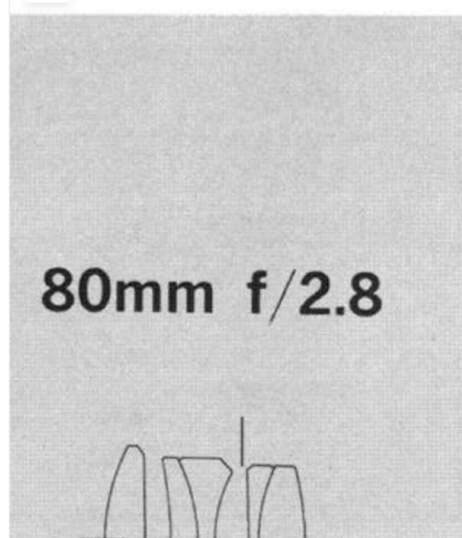
Green zone - the sharpest aperture settings

Orange and red zone - diffraction zone

Interesting thing is that the manufacturing chart goes only up to 9 meters (30 feet) and then up to infinity. On the original chart it starts from 1 foot but since I use this camera only from 2 meters this chart starts from 7 feet.

Why is there no imperial system table? Because it makes me sick 😞
But if you can help me refresh the original table with feet it will be really great!

Original table:



Aperture	Distance (in feet)								
	∞	30	15	10	7	5	4	3	1.5
2.8	102' 7"	23' 4 1/4"	13' 2"	9' 2"	6' 7 1/4"	4' 9 3/4"	3' 10 1/2"	2' 11 1/4"	1' 5 3/4"
	∞	42' 1/4"	17' 5"	11'	7' 5 1/2"	5' 2 1/2"	4' 1 1/2"	3' 3/4"	1' 6 1/4"
4	71' 10 1/2"	21' 4"	12' 6 1/4"	8' 10 1/2"	6' 5 1/4"	4' 8 3/4"	3' 10"	2' 11"	1' 5 5/8"
	∞	50' 9 1/4"	18' 8 3/4"	11' 5 3/4"	7' 8"	5' 3 3/4"	4' 2 1/4"	3' 1 1/4"	1' 6 5/8"
5.6	51' 5"	19' 1 1/2"	11' 9"	8' 5 1/4"	6' 3"	4' 7 1/2"	3' 9 1/4"	2' 10 1/2"	1' 5 3/8"
	∞	70' 4"	20' 9 1/2"	12' 2 1/2"	7' 11 3/4"	5' 5 1/2"	4' 3 1/4"	3' 1 1/2"	1' 6 1/4"
8	36' 3/4"	16' 7"	10' 9 1/4"	7' 11 1/2"	5' 11 3/4"	4' 5 1/4"	3' 8"	2' 10"	1' 5 3/8"
	∞	167' 5 3/4"	24' 11 1/2"	13' 5 3/4"	8' 5 3/4"	5' 8"	4' 4 3/4"	3' 2 1/4"	1' 6 1/4"
11	26' 3 1/2"	14' 2 1/2"	9' 8 3/4"	7' 5"	5' 8"	4' 3 3/4"	3' 6 3/4"	2' 9 1/4"	1' 5 3/8"
	∞	∞	33' 3 3/4"	15' 6 1/2"	9' 2 1/2"	5' 11 3/4"	4' 6 3/4"	3' 3 3/4"	1' 6 1/4"
16	18' 1 3/4"	11' 6"	8' 5"	6' 7 1/2"	5' 2 1/2"	4' 3/4"	3' 4 3/4"	2' 8"	1' 5 3/8"
	∞	∞	76' 3 1/4"	20' 10 1/4"	10' 9 1/2"	6' 6 3/4"	4' 10 1/2"	3' 5"	1' 6 3/8"
22	13' 3 1/4"	9' 4 1/2"	7' 3"	5' 10 3/4"	4' 9 1/4"	3' 9 1/2"	3' 2 3/4"	2' 7"	1' 5 3/8"
	∞	∞	∞	35' 8"	13' 7"	7' 5 1/4"	5' 4"	3' 7 1/2"	1' 7"
32	9' 2 1/4"	7' 2 1/4"	5' 10 3/4"	5"	4' 2 1/4"	3' 5 1/4"	2' 11 1/2"	2' 5"	1' 4 3/8"
	∞	∞	∞	∞	24' 2 1/4"	9' 7 3/4"	6' 3 3/4"	4'	1' 7 1/2"

